Muscles move body parts by contracting and then relaxing. Muscles can pull bones, but they can't push them back to the original position. So they work in pairs of flexors and extensors. The flexor contracts to bend a limb at a joint. Then, when the movement is completed, the flexor relaxes and the extensor contracts to extend or straighten the limb at the same joint. For example, the biceps muscle, in the front of the upper arm, is a flexor, and the triceps, at the back of the upper arm, is an extensor. When you bend your elbow, the biceps contracts. Then the biceps relaxes and the triceps contracts to straighten the elbow.

Points to Remember

- Movement helps to perform necessary functions in an organism. It can be both **voluntary** and **involuntary.**
- Strong muscles and light bones work together to help the birds fly. They fly by flapping their wings.
- Fish swim by forming loops alternately on two sides of the body.
- Snakes slither on the ground by looping sideways. A large number of bones and associated muscles push the body forward.
- The body and legs of cockroaches have hard coverings forming an outer skeleton. The muscles of the breast connected with three pairs of legs and two pairs of wings help the cockroach to walk and fly.
- Earthworms move by alternate extension and contraction of the body using muscles. Tiny bristles on the underside of the body help in gripping the ground.
- Bones and cartilage form the skeleton of the human body. They give the frame and shape to the body and help in movement. It protects the inner organs.
- The skeleton comprises of the skull, the back bone, ribs and the breast bone, shoulder and hip bones, and the bones of hands and legs.
- The bones are moved by alternate contractions and relaxations of two sets of muscles.
- The bone joints are of various kinds depending on the nature of joints and direction of movement they allow.

A-Z GLOSSARY

Antagonist muscle	Muscles that oppose the action of one another.
Appendicular	The arms and legs.
Axial	The trunk and head.
Biceps	Any skeletal muscle having two origins.
Cardiac muscle	Involuntary, striated muscle that constitutes the main tissue of the walls of the heart.
Cartilage	Tough elastic tissue, mostly converted to bone in adults.
Femur	The thigh bone of the human skeleton.
Ligaments	Bands of tough, elastic connective tissue that surround a joint to give support and limit the joint's movement.
Pectoral girdle	The skeletal framework which provides attachment for the forelimbs of or relating to the chest or thorax.
Pelvic girdle	Hips, a foundation for the lower limb.
Skeletal muscle	Voluntary muscles that move bones and produce movement.
Sternum	Long flat bone located in the central part of the chest.
Tendon	Tough band of fibrous connective tissue that usually connects muscle to bone and is capable of withstanding tension.

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